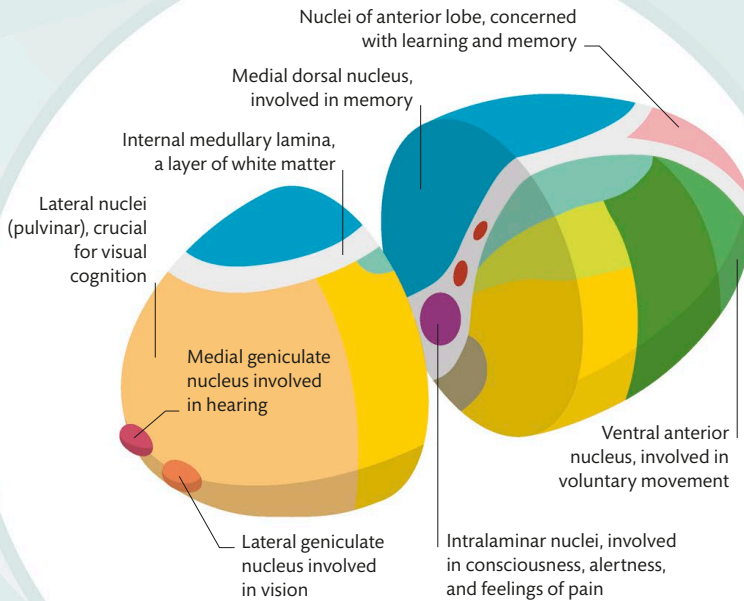




THALAMUS



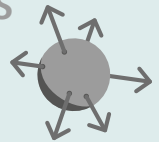
Thalamic nuclei

Investigations into lesser-known nuclei are revealing lots of surprises. For example, the pulvinar nucleus helps the vision centers map out and measure a scene and how we reach out to objects there.

The brain's relay station

It is already well known that the thalamus acts as a relay station for all incoming sensory information (except smell), but more is now being discovered about how it preprocesses this information in a complex and selective manner before it travels to sensory zones in the cortex. The thalamus is also central for the regulation of arousal, and, with its links to the hippocampus, it plays an important role in memory. Deep brain stimulation (see p.185) of the thalamus has been used to treat conditions including tremors.

DESPITE ITS BODY-WIDE EFFECTS, THE SCN CONTAINS ONLY 20,000 NEURONS AND IS SMALLER THAN THIS LETTER O



HAVE ALL THE BRAIN'S PARTS BEEN DISCOVERED?

Not yet. In 2018, improved microscopes uncovered a small region at the brain-spinal cord junction, which was named the endorestiform nucleus.

THE SCN

Located in the hypothalamus, the tiny suprachiasmatic nucleus (SCN) sets the body's circadian rhythm—our 24-hour sleep-wake cycle. This biological clock drives vital homeostatic functions, including body temperature, feeding, and hormone levels. The SCN also coordinates the activities of many organs. Microscopic electrodes or lasers could one day adjust these cycles and patterns.

